LI ZHENXI

Singapore | zli057@e.ntu.edu.sg
Portfolio: portfolio-emsghin2iq-as.a.run.app | linkedin.com/in/zhenxi-li-lzx391 | github.com/zhenxi0901

DevOps & Cloud Engineering Intern | SRE and AI Infrastructure

NTU Computer Science student running two production platforms on Google Cloud: an IoT platform on GKE and an agentic AI platform on Cloud Run. Benchmarks LLM serving on A100 GPUs. ICCV 2025 co-author.

EDUCATION

Nanyang Technological University, Singapore **Jul 2023 - Jun 2027 (expected)**
Bachelor of Computer Science | Exchange: Tsinghua University, Luleå University of Technology

SKILLS

Cloud & Infrastructure: Google Cloud Platform (GKE, Cloud Run, Cloud Build, Cloud SQL, IAM) | AWS (EKS, EC2) | Kubernetes | Docker | Terraform | Pulumi | Linux
CI/CD: GitHub Actions | Jenkins | Google Cloud Build | Git | GitHub
Observability & Reliability: Prometheus | Grafana | PromQL | Cloud Monitoring | Alerting | Load testing
AI Infrastructure: SGLang | A100 GPU | Vertex AI | RAG (Milvus) | Langfuse | LLM benchmarking | PyTorch
Data & Messaging: Kafka | ZooKeeper | PostgreSQL | Redis | MQTT
Languages: Python | Java | Bash | Go | C/C++

WORK EXPERIENCE

Primustech Pte. Ltd., Singapore | DevOps Engineer Intern **May 2026 - Present**

Building IoT platform (Kubernetes on GKE)

- Migrated a live IoT platform from a single Docker Compose VM to **GKE microservices** (Kafka, Cloud SQL, Redis), provisioned with **Terraform** behind Cloud Armor, TLS and autoscaling.
- Own the **Cloud Build CI/CD pipeline** (63 merged PRs): digest-based promotion, PR and Terraform drift gates, approval-gated releases; moved the repository from Bitbucket to GitHub.
- Cut cloud spend **~15%** by tracing a rising monitoring bill to 914 orphaned Kafka topics and shipping an automated cleanup job, plus Cloud SQL and VM right-sizing.
- Built **Grafana dashboards and alerts** for ZooKeeper quorum loss, stalled consumers and crash loops; fixed a ZooKeeper election livelock after node replacement.
- Shipped remote HVAC control over MQTT/RPC with rate limits, a kill switch and simulator tests.

AI agent platform (Cloud Run, Vertex AI)

- Deployed and operated multi-agent (Agno) and **RAG** services on **Cloud Run** in dev, staging and production, backed by Gemini (**Vertex AI**, OpenRouter), **Milvus** vector search and a GPU-served reranker.
- Built its per-environment **Cloud Build CI/CD**, IAM, Secret Manager and **shared-VPC** networking.
- **Benchmarked the LLM stack** with **Langfuse** traces: load and latency tests, per-request token and cost tracking, model comparisons; built a 2,014-device demo building for the agents.

ENGIE South East Asia Pte. Ltd., Singapore | Control Systems Intern **May 2025 - Jan 2026**

- Configured **BMS-room networks and firewalls** across sites, linking DDCs and PLCs to the control room.
- Mapped and validated **800+ BACnet/OPC points** from DDCs and PLCs into SCADA across live sites.
- Programmed DDC and Siemens S7-1200 PLC logic and linked site data to a central monitoring platform.

Shanghai Jiao Tong University, China | Research Intern **May 2024 - Aug 2024**

- Cut NeRF rendering time **40%** and raised image fidelity 30% through CUDA and PyTorch optimisation.

RESEARCH & PUBLICATIONS

Final Year Project: LLM Agent Workload Benchmarking | NTU HyScale Lab **Aug 2026 - Present**

- Contributed a Go benchmark adapter to ARIES, the lab's open-source agent-serving framework, running Toolathlon (109 tasks, 34 MCP servers) in isolated containers.
- **Benchmarked 320 LLM agent runs** on a hosted API and self-hosted SGLang: tool calls take 2.7% of agent time, environment setup 23% of wall clock, and 96% of input tokens hit the prefix cache.
- **Served Qwen3-4B with SGLang** on an A100 over SSH and IAP; jobs held ~7x the GPU time agents used.
- **Load-tested concurrency**: 4 isolated runs per VM gave 3x throughput in 6.4 of 31 GiB memory; root-caused and fixed a Docker race that dropped runs under load.

Publication: PVChat: Personalized Video Chat with One-Shot Learning. ICCV 2025 (co-author).